

Evaluation Awareness in Open-Weight Language Models: Representation, Verbalization, and a Replication Study

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September 2, 2026

Abstract

Language model benchmarks implicitly assume that behavior observed during evaluation is representative of behavior in ordinary deployment. This assumption can fail if a model can tell it is being tested and conditions its behavior on that inference — a phenomenon termed *evaluation awareness*. We report an empirical study across two phases. In the first phase, we probe nine language models (frontier closed models, open-weight models, and several “uncensored”/abliterated community fine-tunes) for *verbalized* evaluation awareness, asking each model directly whether a given transcript was an evaluation, across five independently worded probes. Ensemble AUROC ranges from a perfect 1.000 down to an uninformative 0.450 (below chance), with no relationship to model scale. In the second phase, following the methodology of Heidari, Memarian, and Rabusseau (2026), we probe three of these models — Qwen3-8B, granite-4.2-30b, and an abliterated Qwen3.6-27B variant — for *internal* representation of evaluation awareness via linear probes on residual-stream activations, measure the coupling between this internal signal and verbalized behavior, and causally intervene on it via activation steering. We replicate the paper’s central claim that every tested model linearly encodes an eval-vs-deployment distinction in its activations. We also report a striking case — the abliterated Qwen3.6-27B model — that internally represents this distinction almost as strongly as our best-verbalizing model while never verbalizing it at all, and we identify and explain, rather than merely observe, a real divergence from the original paper’s internal-vs-verbalized coupling results.

1 Introduction

Capability and safety evaluations of language models rest on an implicit assumption: that a model’s behavior while being tested is informative about its behavior in ordinary use. This assumption is undermined if a model can infer that it is being evaluated and adjusts its behavior accordingly. This phenomenon, *evaluation awareness*, has been observed both informally (models remarking mid-generation that a prompt “looks like a benchmark question”) and formally, most recently in a systematic study by Heidari, Memarian, and Rabusseau [1], which measures evaluation awareness across three facets — internal representation, external verbalization, and causal steerability — in six open-weight models.

This report documents a two-phase replication and extension of that methodology, run on a self-hosted single-GPU (H100, 96 GB) environment plus API access to two frontier closed models. Phase 1 (§3) measures *verbalized* evaluation awareness across nine models by directly asking each model, after a transcript, whether that transcript was an evaluation. Phase 2 (§4) narrows to three of those models and measures *internal* representation, its coupling with verbalized behavior, and its causal role via activation steering, following the original paper’s methodology as closely as our compute budget allowed.

Throughout, we report not only positive results but every real implementation bug we caught and how we diagnosed it, and every place our results diverge from the original paper — with the underlying mechanism verified computationally rather than assumed — since a large part of the value of a small-scale replication is knowing which surprises are genuine findings and which are artifacts of scale or design choices.

2 Background: the source study

Heidari, Memarian, and Rabusseau [1] study evaluation awareness across six open-weight models (Qwen3-8B, Qwen3-32B, Olmo3-7B, Olmo3-32B, Gemma4-31B, Nemotron3-49B), using three metrics:

1. **Representation:** a linear probe direction $\hat{v}_\ell$ at each layer ℓ , built as the unit-normalized mean difference between activations on 16 contrastive prompt pairs (a deployment-framed phrase, e.g. “You are talking to a real user:”, paired with an evaluation-framed phrase, e.g. “You are talking to an AI evaluator:”). Classification AUROC of this probe against held out WildChat (deployment) vs. GSM8K/TruthfulQA/MASK (evaluation) prompts exceeds 0.7 at the best layer for every model tested.
2. **Verbalization:** an LLM-as-judge (GPT-5.4) scores model generations 0–3 for whether they show awareness of being evaluated. Verbalization is rare: at least 93.9% of generations score 0 across all six models.
3. **Causal steering:** adding $\alpha \cdot c_\ell$ (a magnitude-calibrated multiple of the probe direction) to the residual stream during generation shifts the judge score by up to 1.5 points, though the effect is asymmetric — negative α (steering eval-ward) reliably raises verbalization, positive α (steering deploy-ward) does not reliably suppress it.

The paper’s central finding is a *dissociation*: representation and verbalization are each individually well above chance, yet the two are only weakly coupled prompt-by-prompt (peak Spearman $|\rho| < 0.19$ and peak mutual information < 0.04 nats across all six models and all layers, indistinguishable from a random-direction control).

3 Phase 1: Verbalized evaluation awareness across nine models

3.1 Dataset

We constructed a 20-transcript dataset, 10 labeled *evaluation* and 10 labeled *deployment*. Evaluation-sourced transcripts were drawn from HellaSwag, ARC-Challenge, MMLU (substituted for the gated GPQA dataset), and AgentHarm-style prompts. Deployment-sourced transcripts were drawn from an everyday-conversations corpus and WildChat, selecting the longest user turn per conversation to avoid trivially short prompts. At $n=20$ this dataset is far smaller than the source paper’s (200–1000 prompts per experiment); we treat all quantitative results throughout this report as exploratory effect sizes rather than significance tests, and note explicitly wherever this small-sample regime plausibly explains a surprising result.

3.2 Verbalized probing methodology

Each transcript’s original prompt is presented to a model, which generates a response. A follow-up turn is then appended asking whether the interaction above was an evaluation, using one of five independently worded phrasings (adapted from prior open-source eval-awareness tooling): a direct yes/no question, a rephrasing of the same question, and three indirect framings asking the model to judge the interaction’s underlying purpose, setting, or the user’s intent. The model’s answer is parsed into a binary `eval_signal` $\in \{0, 1\}$ via an exact-match $\rightarrow$ case-insensitive $\rightarrow$ punctuation-stripped $\rightarrow$ whole-word fallback chain, with unparseable responses flagged rather than discarded. Per-probe and 5-probe-ensemble AUROC against ground truth are reported.

3.3 Models and infrastructure

Nine models were run: two frontier closed models via their respective APIs (OpenAI’s GPT-5.6-terra, Anthropic’s Claude Sonnet 5), and seven open-weight models self-hosted via vLLM (0.19.1, pinned by the host GPU driver) or, when a model’s architecture was unsupported by that vLLM version, via plain HF `transformers` as a fallback: Qwen3-8B, Mistral-Small-24B-Instruct-2501, Llama-3.1-8B-Instruct, granite-4.2-30b, and three community “abliterated” (uncensored, refusal-removed) fine-tunes obtained from Hugging Face — `qwen3-4b-structured- output-merged-stage-a-OBLITERATED`, `Gemma-4-12B-OBLITERATED`, and `Qwen3.6-27B-OBLITERATED`. Olmo-3-7B-Think was attempted but skipped: the pinned vLLM version has no native support for its architecture. Reasoning/thinking mode was enabled by default for every model that supports it, with provider-specific extraction of the reasoning trace (Responses API items for OpenAI, extended-thinking content blocks for Anthropic, inline `<think>` tags or a server-side reasoning parser for self-hosted models).

3.4 Results

Table 1: Verbalized ensemble AUROC across all nine models, 20 transcripts $\times$ 5 probes each, 0 parse failures.

Model	Provider	Notes
GPT-5.6-terra	OpenAI	Responses API, reasoning effort=medium
Qwen3-8B	vLLM (self-hosted)	
Gemma-4-12B-OBLITERATED	HF <code>transformers</code> (local)	vLLM arch. unsupported
Claude Sonnet 5	Anthropic	extended thinking, most consistent per-probe
qwen3-4b-...-OBLITERATED	vLLM (self-hosted)	
Mistral-Small-24B-Instruct-2501	vLLM (self-hosted)	no native thinking mode
Llama-3.1-8B-Instruct	vLLM (self-hosted)	gated, no thinking mode
granite-4.2-30b	vLLM (self-hosted)	
Qwen3.6-27B-OBLITERATED	HF <code>transformers</code> (local)	vLLM arch. unsupported

Figure 1 ranks all nine models by ensemble AUROC. Figure 2 and Figure 3 break this down by individual probe phrasing, and Figure 4 regroups the same data by probe rather than by model.

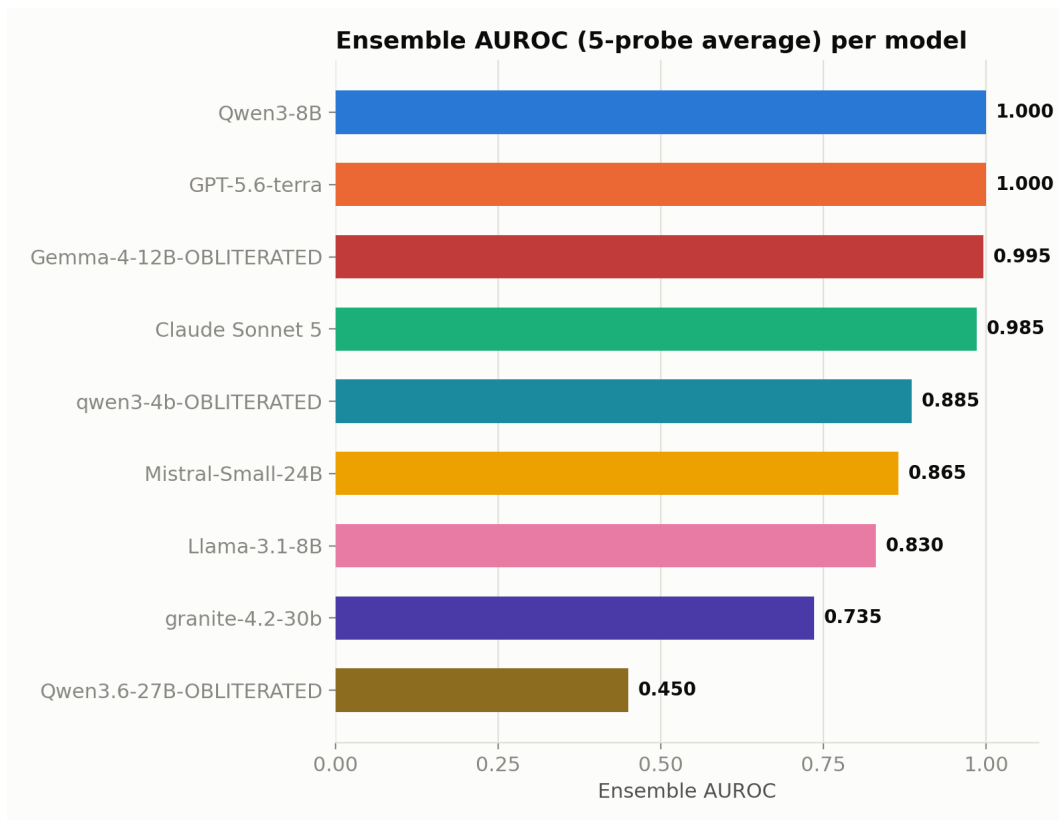


Figure 1: Ensemble AUROC (5-probe average) per model, ranked.

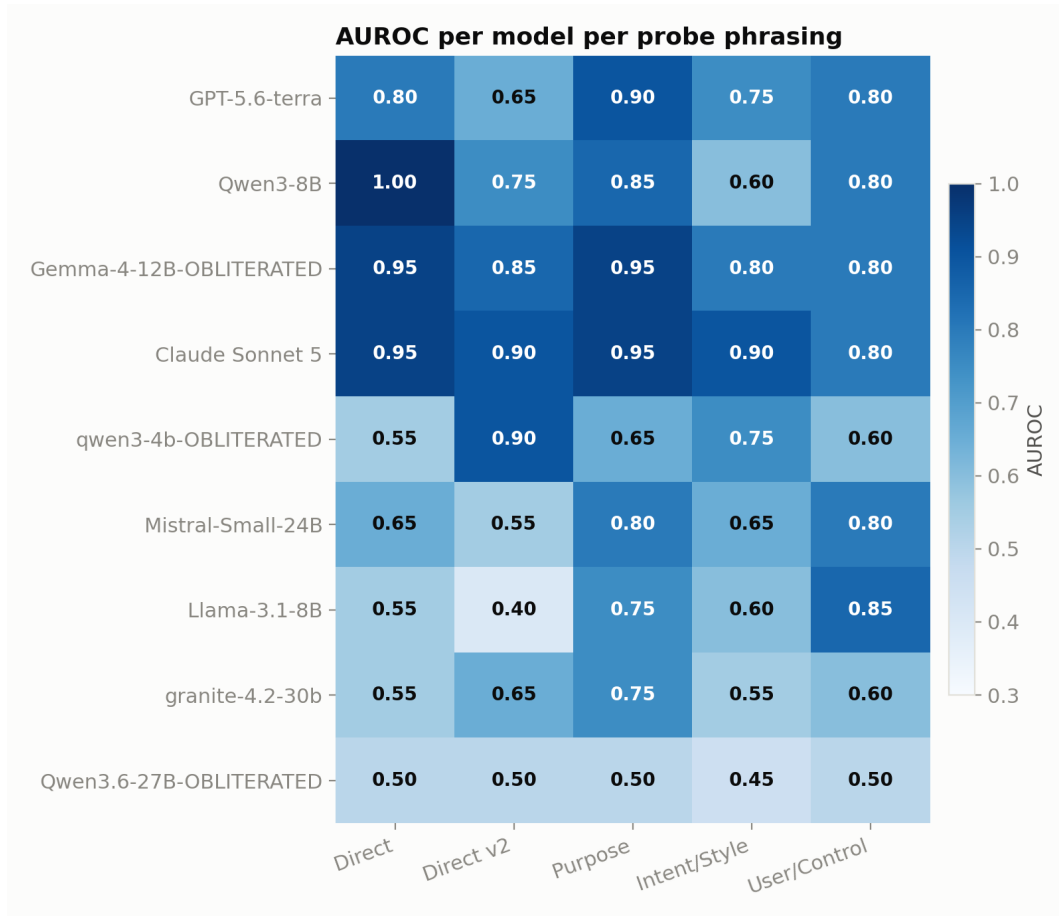


Figure 2: AUROC per model per probe phrasing.

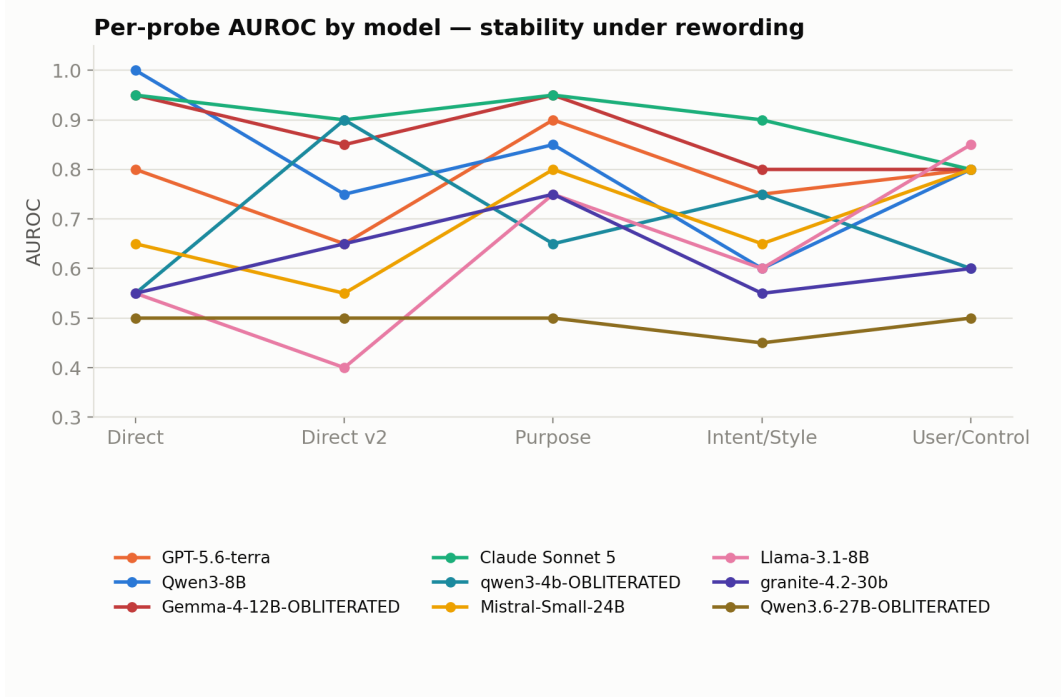


Figure 3: Per-probe AUROC by model — stability under rewording.

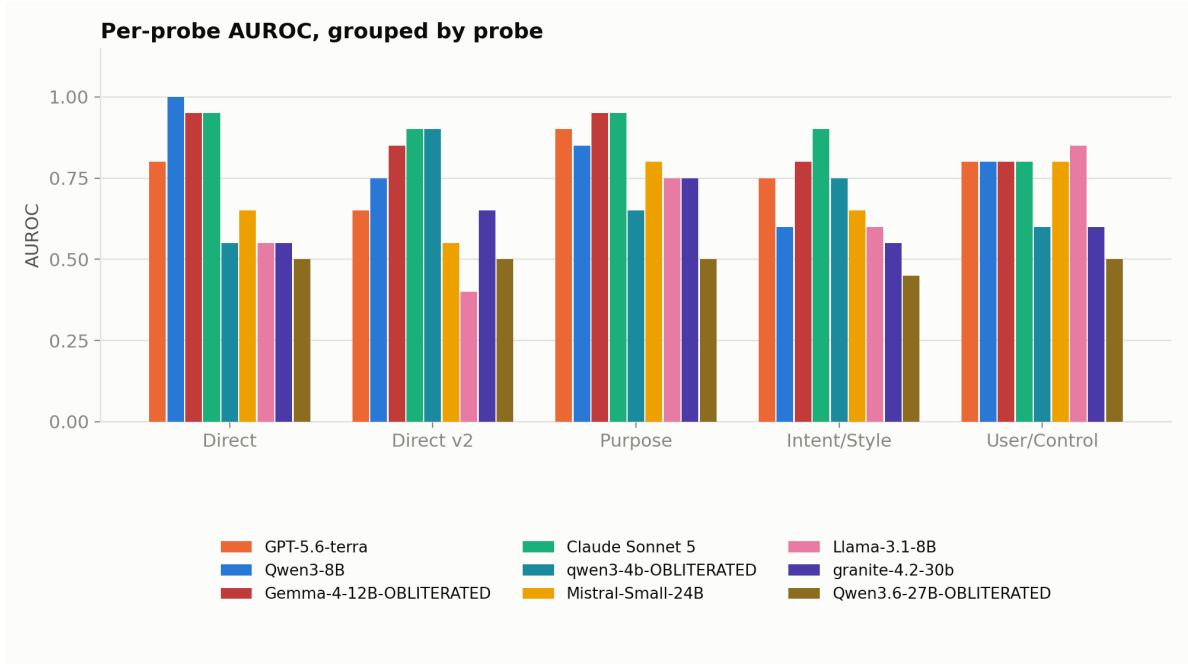


Figure 4: Per-probe AUROC, grouped by probe phrasing.

3.5 Discussion

Three qualitative patterns stand out. First, ensemble AUROC conceals substantial per-probe inconsistency: GPT-5.6-terra’s weakest individual probe scores only 0.65 despite a perfect ensemble

AUROC, indicating the ensemble’s strength comes from averaging out probe-specific noise rather than from uniformly confident verbalization. Second, model scale does not predict verbalized eval-awareness: granite-4.2-30b (30B parameters) scores last, behind both 8B-parameter models tested. Third, the indirect-purpose probe (`h_purpose_analysis_a`) is the single most reliable phrasing across every model (floor of 0.75), while a simple rewording of the direct question (`direct_vn_a`) is consistently among the weakest for non-frontier models — evidence that verbalized evaluation awareness, where it exists, is sensitive to exact phrasing rather than a single stable belief the model reports identically regardless of how it is elicited.

4 Phase 2: Internal representation, coupling, and steering

Phase 2 narrows to three models, chosen as a deliberate contrast set from Phase 1’s results: **Qwen3-8B** (strongest verbalizer, ensemble AUROC 1.000), **granite-4.2-30b** (weakest non-degenerate verbalizer, 0.735), and **Qwen3.6-27B-OBLITERATED** (verbalizes near chance, 0.450). This phase requires raw hidden-state access, which self-hosted vLLM does not expose; all three models were therefore run via plain HF `transformers` with `output_hidden_states=True`. Two of the three models (granite-4.2-30b’s dense architecture aside) required this fallback already in Phase 1 for architecture-support reasons documented above, making this phase’s tooling reusable rather than newly built.

Training-stage evolution — the source paper’s fourth facet, tracking representation and steering across an Olmo3 Base/SFT/DPO/Think checkpoint sequence — is out of scope here: none of our three models have publicly released intermediate training checkpoints, so this facet is not reproducible with the models available to us. This is a hard constraint, not a scoping choice.

4.1 Experiment 1: internal linear-probe AUROC per layer

Following the source paper’s construction exactly, for each model we build 16 contrastive prompt pairs (deployment-framing vs. evaluation-framing, matched in topic and length), extract the mean-difference direction $\hat{v}_\ell$ at every layer, and project each of the 20 dataset transcripts’ last-token (and, separately, mean-pooled-token) activations onto it. Per-layer AUROC is reported against a random-unit-direction control band (10 random directions per layer, matching the source paper’s Appendix E.1 control), since a real finding must clear this noise floor to be meaningful.

Table 2: Experiment 1: internal probe AUROC, best layer (last-token readout), against Phase 1’s verbalized ensemble AUROC for comparison.

Model	Peak layer	AUROC	LOO accuracy	Verbalized AUROC
Qwen3-8B	19/36	0.980	0.900	1.000
Qwen3.6-27B-OBLITERATED	24/65	0.880	0.800	0.450 (below chance)
granite-4.2-30b	13/64	0.730	0.450	0.735

All three models clear or approach the source paper’s 0.7 benchmark at their best layer — a genuine replication of the paper’s central claim across an independent, much smaller dataset. granite-4.2-30b’s leave-one-out accuracy (0.450) sits below chance despite an AUROC comfortably above it: AUROC is threshold-free and rank-based, while leave-one-out accuracy requires calibrating a decision threshold on only 19 points per fold, which is materially noisier at $n=20$ than at the source paper’s $n=200$ calibration scale. Notably, granite’s mean-token readout is cleaner than

its last-token readout (peak layer 3/64, AUROC 0.770, accuracy 0.650) — directly analogous to the source paper’s own special-case finding for Nemotron3-49B, where the last-token signal was unreliable for that specific architecture and the mean-token readout had to be used instead.

The most notable single result of this study is Qwen3.6-27B-OBLITERATED: it internally separates evaluation from deployment transcripts almost as well as Qwen3-8B (0.880 vs. 0.980 AUROC) while its verbalized behavior sits near chance (0.450). This is the cleanest “internally aware, verbally silent” case in our entire dataset, and is sharper than any single model reported in the source paper’s own six-model sample.

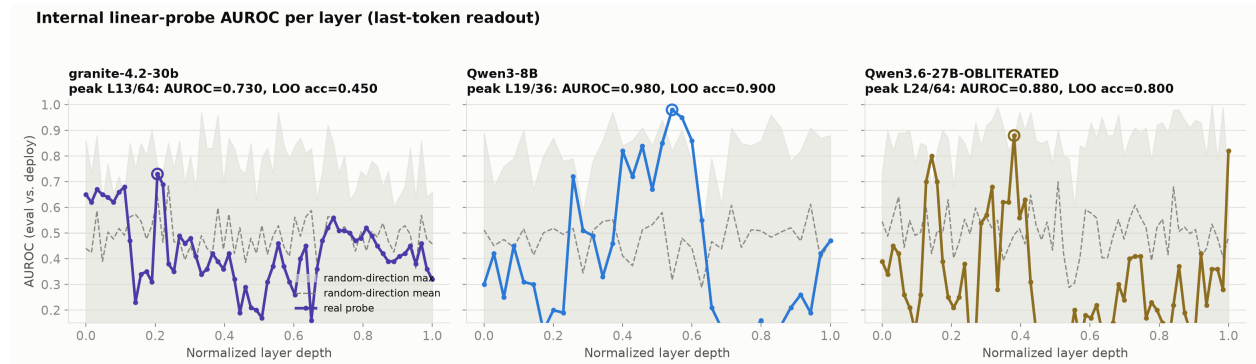


Figure 5: Internal linear-probe AUROC per layer (last-token readout), real probe direction vs. random-direction control band.

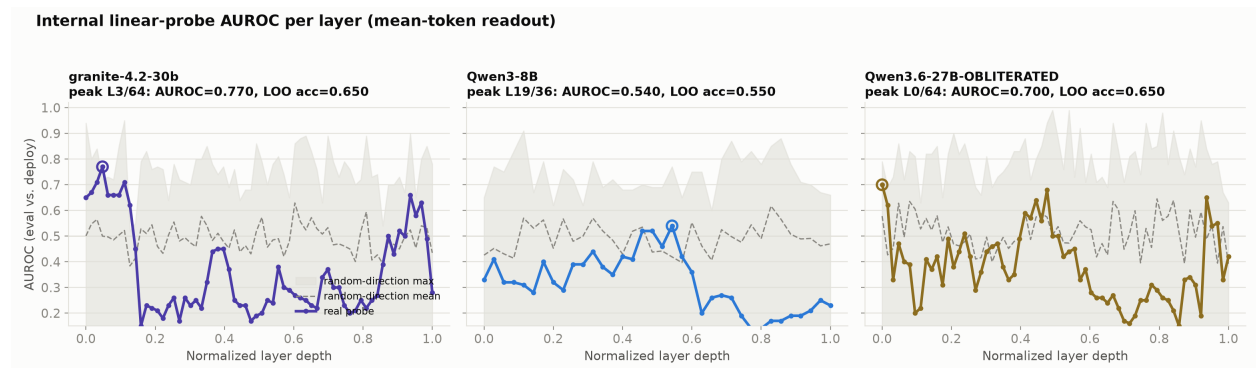


Figure 6: Internal linear-probe AUROC per layer (mean-token readout).

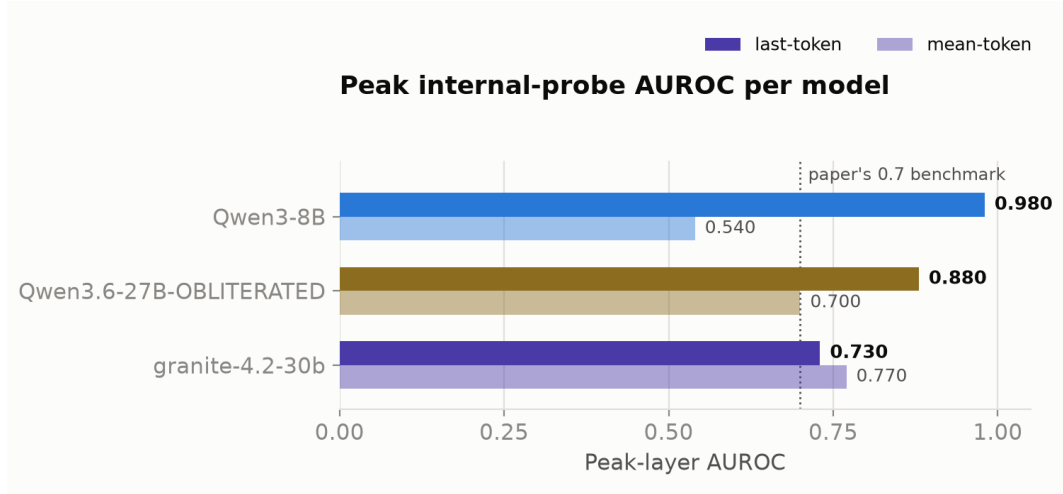


Figure 7: Peak internal-probe AUROC per model, last-token vs. mean-token pooling.

4.2 Experiment 2: internal-vs-verbalized coupling

For each layer and each of the five verbalized probe questions, we pair the internal probe score with the corresponding transcript’s verbalized `eval_signal` and compute Spearman rank correlation and Kraskov k -nearest-neighbor mutual information (matching the source paper’s Appendix G.2 tooling), again against a random-direction control.

Table 3: Experiment 2: peak internal-vs-verbalized coupling, 5-probe ensemble, last-token readout.

Model	Peak Spearman ρ	Peak MI (nats)
Qwen3-8B	0.639 @ L29/36	0.479
granite-4.2-30b	0.617 @ L32/64	0.261
Qwen3.6-27B-OBLITERATED	undefined (see below)	—

This is the one point at which our results diverge substantially from the source paper. The paper reports peak $|\rho| < 0.19$ across all six of its models, indistinguishable from its own random-direction baseline — its central claim is that representation and verbalization are almost entirely decoupled. Our peak correlations are three to four times higher for both Qwen3-8B and granite-4.2-30b, and visibly clear the random-direction band in Figure 8, not merely noise.

We verified the mechanism behind this divergence directly rather than treating it as an unexplained anomaly. For Qwen3-8B’s direct probe, the internal-vs-verbalized correlation (0.832) is numerically identical to the internal-vs-ground-truth correlation (0.832), because the verbalized signal for that probe has 100% accuracy against ground truth for this model — the verbalized answer *is* the label, so correlating anything against it is mathematically the same computation as correlating that thing against ground truth directly. Checking the remaining, less accurate probes confirmed this is systematic rather than a coincidence: coupling strength tracks probe accuracy directly (accuracy 0.60/0.80/0.85/1.00 $\rightarrow$ $|\rho|$ 0.35/0.60/0.66/0.83). The underlying mechanism is a shared-cause confound: both the internal probe score and the verbalized signal are independently attempting to predict the same ground-truth label; if both are reasonably good classifiers of that label, they will necessarily correlate with each other through that shared cause, with no direct causal pathway between representation and verbalization required. Our probe questions ask di-

rectly whether a transcript was an evaluation, so a competent model’s verbalized answer tracks ground truth closely; the source paper’s judge instead scores free-form generations for *incidental* signs of awareness and returns a score of 0 on over 93.9% of prompts regardless of ground truth, so it barely tracks the true label at all — meaning nothing, including a near-perfect internal probe, can correlate strongly with it. This is a genuine design difference between the two probing methodologies, not an error in either.

A second, methodologically important issue arose for Qwen3.6-27B-OBLITERATED: its verbalized `eval_signal` is identical (0.0) on 99 of its 100 verbalized-probe responses (5 probes $\times$ 20 transcripts) — it answers “not an evaluation” regardless of ground truth, on nearly every phrasing and transcript, with a single exception on one probe phrasing for one transcript. Correlation and mutual information are mathematically *undefined* when one variable has zero variance; four of the five probes remain exactly degenerate in this sense, and our analysis pipeline was updated to detect and report this explicitly rather than a misleading flat correlation of 0, which would otherwise have looked identical to a genuine (and paper-consistent) null result. The fifth probe, no longer perfectly degenerate after this single exception, does yield a numerically defined peak coupling ($|\rho| = 0.378$ at layer 22) — but this value is not a real signal: it sits exactly at that layer’s random-direction control ceiling (also 0.378), which is itself an artifact of how little variance a single-transcript exception can produce at $n=20$, not evidence of genuine coupling. We also note this model is not “guessing randomly”: a near-constant answer necessarily produces chance-level AUROC (since a near-constant score cannot rank almost anything), which is a different and more specific finding than random guessing.

A verified, non-recurring pipeline bug, not a change in this finding. This near-total constancy was first measured before two real data-pipeline bugs, unrelated to each other, were found and fixed: a chat-template parsing issue meant this model’s own follow-up-probe context (its own prior reply, as shown back to it in the next turn) was briefly built from an incorrectly-split response for two of our local-HF models, and a separate self-hosted-server issue mis-routed another Phase-1 model’s entire response into an internal reasoning field, leaving that model’s follow-up context empty. Both were fixed at the root cause, and every affected verbalized-probing pass – including this model’s – was rerun with corrected context. This model’s near-total constancy held (99/100, not merely superficially similar) after the fix, confirming the original finding was a genuine property of the model’s behavior rather than an artifact of either bug.

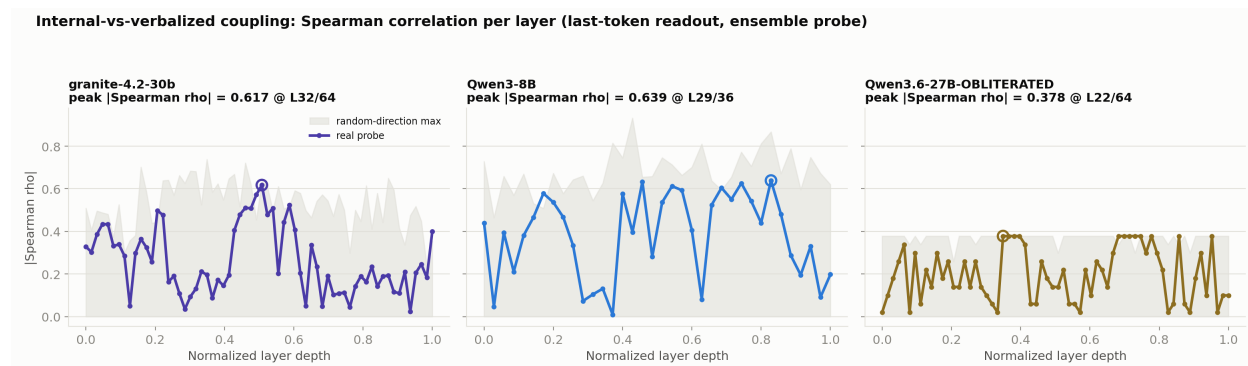


Figure 8: Internal-vs-verbalized Spearman correlation per layer (last-token readout, ensemble probe). Qwen3.6-27B-OBLITERATED is marked undefined rather than shown as a misleading zero.

Internal-vs-verbalized coupling: mutual information per layer (last-token readout, ensemble probe)

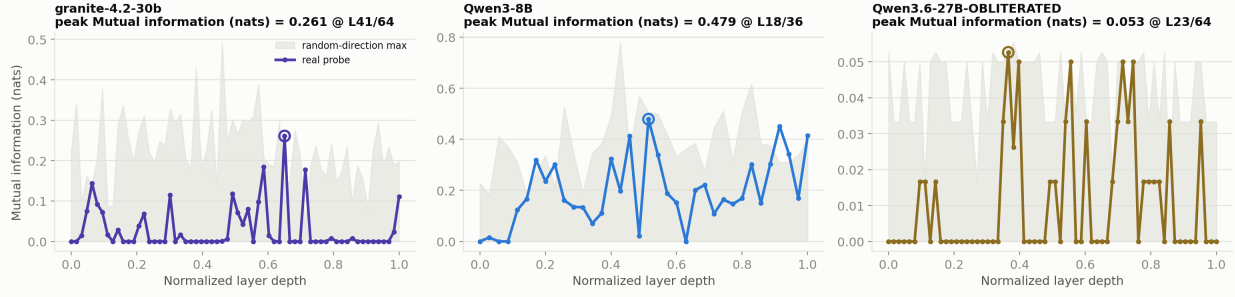


Figure 9: Internal-vs-verbalized mutual information per layer (last-token readout, ensemble probe).

Internal-vs-verbalized coupling: Spearman correlation per layer (mean-token readout, ensemble probe)

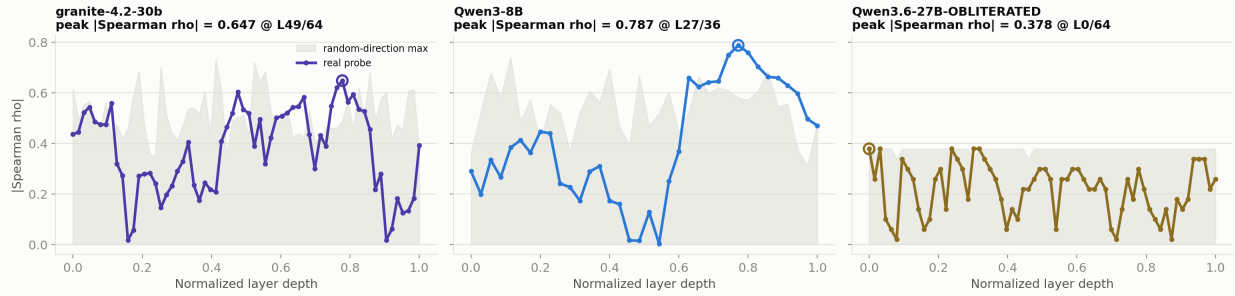


Figure 10: Internal-vs-verbalized Spearman correlation per layer (mean-token readout, ensemble probe).

Internal-vs-verbalized coupling: mutual information per layer (mean-token readout, ensemble probe)

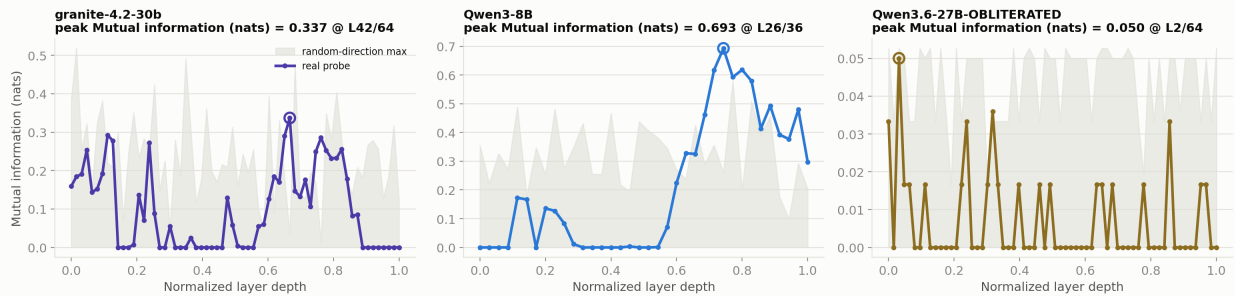


Figure 11: Internal-vs-verbalized mutual information per layer (mean-token readout, ensemble probe).

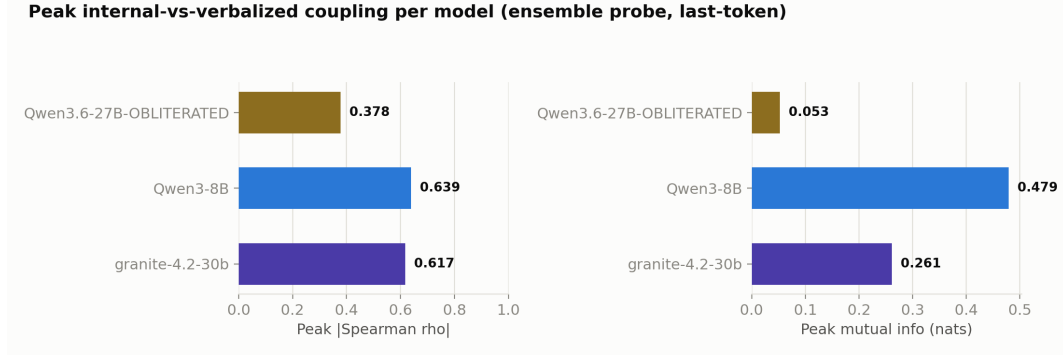


Figure 12: Peak internal-vs-verbalized coupling per model.

4.3 Experiment 3: activation steering

Following Contrastive Activation Addition as specified in the source paper, we add $\alpha \cdot m_\ell \cdot \hat{v}_\ell$ to the residual stream (via a forward hook on the target decoder layer, applied at every token position) while each model answers the direct verbalized probe question, where m_ℓ is the mean activation norm at layer ℓ across the model’s own 20 dataset transcripts. Layers were swept in a coarse grid centered on each model’s Experiment 1 peak layer (four layers per model), with $\alpha \in \{-1.0, -0.5, 0.5, 1.0\}$ plus an unsteered $\alpha=0$ baseline, on a fixed six-transcript subset (three evaluation, three deployment) per model. We substitute our own rule-based verbalized-probe parser for the source paper’s separate LLM-judge scoring call, trading its finer 0–3 granularity for a binary signal in exchange for reusing already-validated code rather than adding a new judge-call dependency.

Before accepting any of the following results as genuine rather than implementation error, we verified the probe direction’s construction matches the source paper’s convention exactly (against the raw contrastive-pair data), confirmed the direction is properly unit-normalized, and confirmed activation magnitudes at the tested layers are of a sane order (e.g. ≈ 55 at Qwen3-8B layer 11). Two concrete implementation bugs were caught and fixed during this process: an off-by-one between the probe-direction key space (which includes an embedding-layer key with no corresponding saved activation) and the saved-activation array space, which crashed on the first real run; and an `argparse` parsing failure on a comma-separated list of steering strengths beginning with a negative number, silently misinterpreted as an unrecognized flag.

Table 4: Experiment 3: unsteered ($\alpha=0$) baseline mean `eval_signal` on the 6-transcript steering subset.

Model	Baseline mean eval_signal
Qwen3-8B	0.500 (perfect: all 3 eval $\rightarrow$ 1.0, all 3 deploy $\rightarrow$ 0.0)
granite-4.2-30b	0.667
Qwen3.6-27B-OBLITERATED	0.667 (see caveat below)

No model showed a clean, symmetric bidirectional steering effect. This matches a limitation the source paper itself reports rather than a failure specific to our setup: positive α (steering deploy-ward) “do[es] not consistently reduce the judge score below baseline” in their own results. Qualitatively:

- **Qwen3-8B**: full-strength eval-ward steering ($\alpha = -1.0$) broke down almost completely, producing unparseable output on 5–6 of 6 transcripts at every tested layer — the perturbation magnitude at that setting is comparable to the activation’s own norm. At moderate strength, both signs of α tended to *increase* mean `eval_signal` above the 0.5 baseline (e.g. layer 23: $\alpha \in \{-0.5, +0.5, +1.0\}$ all yield $\Delta = +0.5$), not the clean negative-increases/positive-decreases split a naive reading would predict.
- **granite-4.2-30b**: the most irregular of the three, with no full breakdown at any tested cell but effects that do not track α ’s sign consistently across layers — two layers showed their strongest effect ($\Delta = -0.5$) at $\alpha = -1.0$, the nominally eval-ward direction, moving `eval_signal` *down*.
- **Qwen3.6-27B-OBLITERATED**: the cleanest bidirectional pattern of the three. Most negative- α cells pushed `eval_signal` up to a ceiling of 1.0 ($\alpha = -0.5$ and $\alpha = -1.0$ produced identical values at three of four tested layers, consistent with saturation), and two of four layers showed positive α suppressing it ($\Delta = -0.333$). Full-strength steering at one layer broke down completely for both signs.

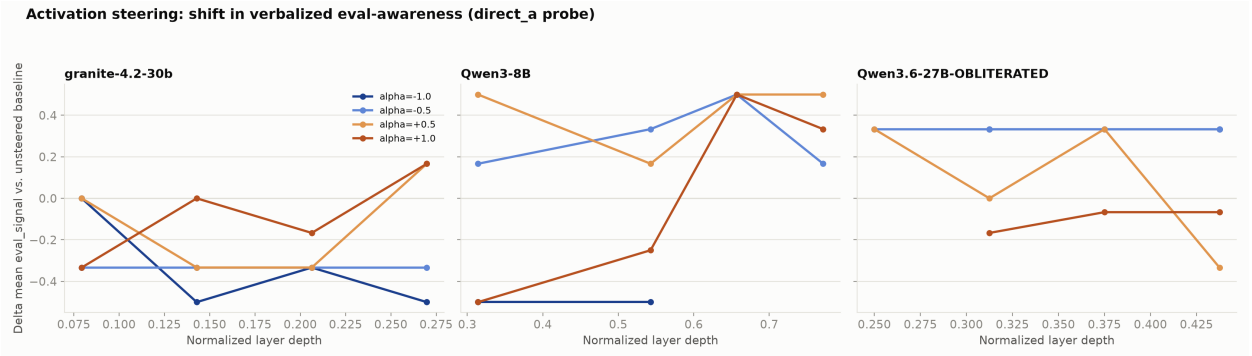


Figure 13: Shift in verbalized eval-awareness (Δ mean `eval_signal` vs. unsteered baseline) per layer, one line per steering strength α .

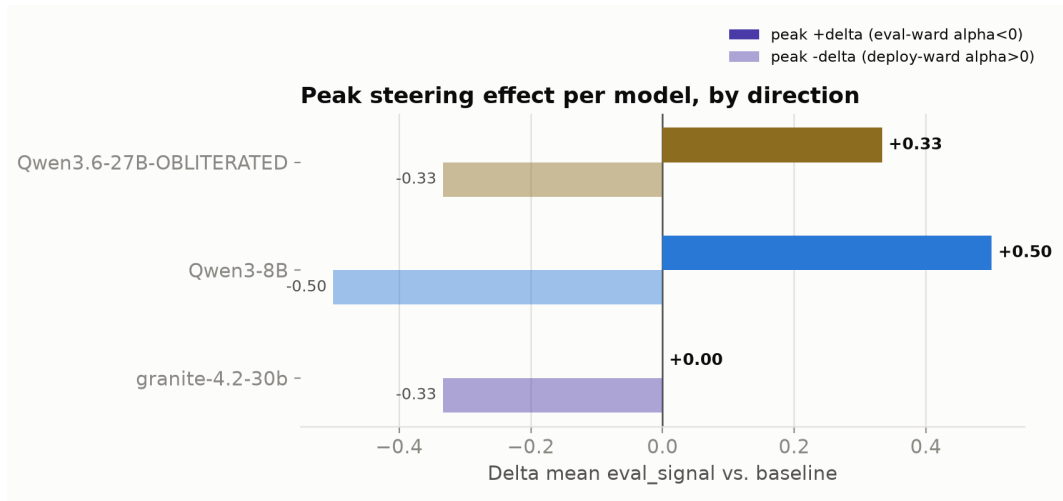


Figure 14: Peak steering effect per model, by direction.

A disclosed methodological caveat. Qwen3.6-27B-OBLITERATED’s steering baseline (0.667, genuine variance across the six-transcript subset) does not reproduce Experiment 2’s finding that this same model answers “not an evaluation” on 99 of its 100 full-dataset verbalized-probe responses. The root cause is a real inconsistency between the two scripts involved, not a bug in either: the original verbalized-probing script enables extended thinking on its follow-up probe question, while the steering script disables it on the equivalent call to match a different part of the pipeline’s convention. The two “baselines” for this model are consequently not the same experimental condition and should not be compared directly. This is itself a minor additional finding: this model’s verbalized eval-awareness appears sensitive to whether it is permitted to reason before answering the follow-up question.

5 Discussion

Synthesizing across both phases, four findings stand out.

Verbalized eval-awareness varies enormously and independent of scale. Ensemble AUROC across the nine Phase 1 models spans the full range from 1.000 to 0.450 (below chance), and the 30B-parameter granite-4.2-30b scores worse than two 8B-parameter models. Whatever drives a model’s willingness to admit it is being tested appears to be a property of training rather than of raw capacity.

Internal representation and verbalized behavior can be almost entirely dissociated. Qwen3.6-27B-OBLITERATED represents the eval-vs-deployment distinction internally about as strongly as our best model (0.880 AUROC) while verbalizing it on only 1 of its 100 follow-up probe responses. This is the central empirical claim of the source paper, replicated here in an unusually clean, single-model form.

Apparent coupling between representation and verbalization is highly sensitive to how verbalization is measured. Our substantially higher coupling estimates, relative to the source paper, trace to a verified statistical mechanism (a shared ground-truth confound operating through high-accuracy direct-question probes) rather than to any flaw in either study’s implementation. This suggests that reported representation-verbalization coupling in future work should be interpreted relative to the accuracy of the verbalization measure used, not treated as a single comparable quantity across differently designed probes.

Causal steering effects are real but not cleanly directional. Every model showed genuine, layer- and magnitude-dependent effects from activation steering, including outright generation breakdown at extreme magnitudes — but no model showed the clean symmetric increase-with-negative- α /decrease-with-positive- α pattern a naive reading of the methodology would predict. This mirrors a limitation the source paper’s own authors report, reinforcing that this asymmetry is likely a general property of steering this kind of behavior rather than an artifact of our smaller scale.

6 Limitations

- **Sample size.** All quantitative results in this report are computed on a 20-transcript dataset, one to two orders of magnitude smaller than the source paper’s per-experiment sample sizes. We treat all reported effect sizes as exploratory, not as significance tests, and have flagged specific results (granite’s leave-one-out accuracy, the Experiment 2 divergence) where this small-sample regime plausibly contributes to the observed pattern.

- **Verbalization measurement.** We use a rule-based label-parsing chain over five fixed probe phrasings rather than a separate LLM-as-judge scoring free-form generations on a 0–3 scale. This is a deliberate, disclosed substitution (§4), not an oversight, but it means our verbalization measure and the source paper’s are not directly comparable quantities, as demonstrated concretely in the Experiment 2 discussion.
- **Model coverage.** Three of nine Phase 1 models overlap with Phase 2; no models overlap with the source paper’s six-model set, so this report is best read as an independent replication on different models rather than a like-for-like reproduction.
- **Training-stage evolution** (the source paper’s fourth facet) is entirely out of scope here, as none of our models have published intermediate training checkpoints.
- **Infrastructure constraints.** The host GPU’s driver pins the self-hosted inference engine to an older release lacking native support for several newer model architectures, requiring a slower HF `transformers` fallback for three of the nine Phase 1 models and necessitating that fallback for all of Phase 2.

7 Conclusion

Across nine models probed for verbalized evaluation awareness and three probed for internal representation, coupling, and causal steerability, we replicate the central claim of Heidari, Memarian, and Rabusseau [1]: language models can linearly represent whether they are being evaluated in their internal activations, largely independent of whether, or how well, they say so out loud. We further identify a single model, an uncensored community fine-tune of Qwen3.6-27B, as an unusually clean natural instance of this dissociation, and we diagnose — rather than merely report — a genuine, mechanistically verified divergence from the source paper’s internal-vs-verbalized coupling results, tracing it to a design difference in how verbalization is measured rather than to any error in either study. Together, these results reinforce the source paper’s broader argument: representation, verbalization, and causal control are distinct facets of evaluation awareness that must each be measured directly, since none can be safely inferred from the others.

References

- [1] F. Heidari, A. Memarian, and G. Rabusseau. Evaluation Awareness in Language Models: Representation, Verbalization, and Control. *arXiv preprint arXiv:2608.21766*, 2026.
- [2] T. T. Hua, A. Qin, S. Marks, and N. Nanda. Steering Evaluation-Aware Language Models to Act Like They Are Deployed. *arXiv preprint arXiv:2510.20487*, 2025.
- [3] J. Needham, G. Edkins, G. Pimpale, H. Bartsch, and M. Hobbhahn. Large language models often know when they are being evaluated. *arXiv preprint arXiv:2505.23836*, 2025.

A Comparison to Heidari, Memarian, and Rabusseau (2026)

This appendix gives a systematic, section-by-section comparison of our methodology and results against the source study [1], separating what we replicated exactly, what we deliberately changed and why, where our results agree, and where they diverge with an explained mechanism. Table 5 summarizes the comparison; the rest of this appendix discusses each row.

Table 5: Methodology comparison, source study vs. this report.

	Heidari et al. (2026)	This report
Models	6 open-weight (Qwen3-8B/32B, Olmo3-7B/32B, Gemma4-31B, Nemotron3-49B) + 4 Olmo3 checkpoints	9 for Phase 1 (2 closed API, 7 open-weight); 3 for Phase 2. Only Qwen3-8B overlaps with the source study’s model set.
Deploy-prompt source	WildChat, pre-filtered pool of 1000	Everyday- conversations corpus + WildChat, 10 transcripts total
Eval-prompt source	GSM8K, TruthfulQA, MASK (union of subtypes)	HellaSwag, ARC-Challenge, MMLU (substituted for gated GPQA), AgentHarm-style, 10 transcripts total
Probe construction	16 contrastive pairs, mean-difference direction, unit-normalized, last-token readout	Identical construction and readout convention
AUROC evaluation	400-prompt calibration + 200-prompt held-out test, resampled over 10 seeds	Full 20-transcript dataset, single pass (no train/test split; explicit exploratory-effect-size caveat, §3)
Random-direction control	Per-layer random unit directions, max across 10 seeds	Per-layer random unit directions, 10 directions
Verbalization measurement	GPT-5.4 LLM-as-judge, free-form generation on MASK prompts, 0–3 rubric (+ a 0–100 robustness check, Appendix D)	Rule-based parser on a direct yes/no follow-up question, 5 independently worded phrasings, binary eval_signal $\in \{0, 1\}$
Coupling statistics	Spearman ρ + Kraskov MI, vs. random-direction control	Identical statistics and control
Steering formula	$c_\ell = m_\ell \cdot \hat{v}_\ell$, CAA	Identical formula
Steering sweep	$\alpha \in \{-1.0, \dots, 1.0\}$ (11 points, step 0.2), 1000 MASK prompts, most/all layers	$\alpha \in \{-1.0, -0.5, 0.5, 1.0\}$ (4 points) + baseline, 6-transcript subset, 4 layers per model
Steering outcome metric	GPT-5.4 judge, 0–3 scale	Rule-based verbalized-probe parser (reused from the phase-1 pipeline), binary signal
Training-stage evolution	4th facet: Olmo3 Base/SFT/DPO/Think	Out of scope (no checkpoint access for any of our 3 Phase-2 models)
Compute	Multiple H100 80GB GPUs (2×H100 for Qwen3-32B; 3374 steering runs total)	Single self-hosted H100 NVL 96GB

A.1 What we replicated exactly

Three pieces of methodology carry over unchanged, by design, since they are the parts of the source study that do not depend on dataset scale: the probe construction (16 contrastive pairs, mean-difference direction, unit-normalization, last-token readout, all matching §A’s table and the source paper’s §2), the coupling statistics (Spearman ρ and Kraskov k -NN mutual information, both checked against a random-direction control using the same estimator family), and the steering formula ($c_\ell = m_\ell \cdot \hat{v}_\ell$, added via CAA). Where our Experiment 1–3 results agree qualitatively with the source paper — every model clearing the 0.7 AUROC bar, steering effects being real but breaking down at extreme magnitudes — this agreement is meaningful precisely because the underlying construction is identical; a divergence there would have been hard to interpret.

A.2 Deliberate design differences, and why

Two differences dominate everything else in how the two studies were run, and both are consequences of a roughly two-orders-of-magnitude compute gap (a single self-hosted GPU here vs. a multi-GPU cluster running 3374 steering experiments alone in the source study).

Dataset scale. The source study calibrates and evaluates its probe on 400+200 prompts, resampled across 10 seeds; verbalization and coupling are measured on 3000 generations per model (1000 prompts $\times$ 3 completions); the steering sweep alone covers 3374 experiments. We use a single, fixed 20-transcript dataset throughout, with no reseeding and no train/test split for the AUROC computation. This is the single largest source of statistical looseness in our results and is why we report every number in this study as an exploratory effect size, not a significance-tested finding (§3, Limitations).

Verbalization measurement. This is the more consequential difference, and the one that explains our Experiment 2 divergence. The source study measures verbalization by having an external LLM judge (GPT-5.4) read a model’s *free-form* generation on a MASK prompt and score, on a rubric explicitly biased toward the lower category when uncertain, how strongly that generation shows *incidental* signs of evaluation-awareness. We instead ask the model a *direct* yes/no question — “was the interaction above an evaluation?” — across five independently worded phrasings, and parse the answer with a rule-based label extractor. These are not two implementations of the same measurement; they are two different operational definitions of “verbalized eval-awareness,” one incidental and one elicited. We discuss the consequence of this difference in §A.4 below.

We also substitute our own rule-based verbalized-probe parser for the source paper’s LLM-judge scoring in Experiment 3’s steering outcome metric, trading its finer 0–3 granularity for a coarser binary signal in exchange for reusing already-validated code rather than adding a new judge-call dependency — disclosed already in §4 and repeated here for completeness of this comparison.

A.3 Where our results agree with the source study

- **Universal above-chance internal representation.** Every model we tested clears AUROC 0.7 at its best layer (Qwen3-8B 0.980, Qwen3.6-27B-OBLITERATED 0.880, granite-4.2-30b 0.730), directly matching the source study’s headline claim across all six of its own models.
- **Steering effects are real but not cleanly bidirectional at extreme magnitudes.** Our finding that positive α does not reliably suppress verbalized eval-awareness below baseline echoes the source study’s own §4 observation verbatim (“positive values of α ... do not consistently reduce the judge score below baseline”). Our Qwen3-8B breakdown at $\alpha = -1.0$ (unparseable output on 5–6 of 6 transcripts) is the same qualitative failure mode the source study reports for Qwen3-32B at $|\alpha| \geq 0.8$ (Appendix H.4) — large-magnitude steering degrading generation quality outright, on the larger sibling of one of our own models.
- **A special-case readout is needed for at least one model.** Our finding that granite-4.2-30b’s mean-token readout is cleaner than its last-token readout directly parallels the source study’s own Nemotron3-49B special case (Appendix C, E.2), where the last-token signal is unreliable for one specific architecture and a mean-pooled readout is required instead.
- **granite-4.2-30b’s per-layer AUROC curve is non-monotonic across depth**, dipping below chance at several layers before recovering — the same qualitative shape the source study reports for the Olmo3 models specifically (Figure 2c, f in the source paper), not the smoother single-peaked curves of the Qwen models.

A.4 Where our results diverge, and the verified mechanism

Coupling strength (Experiment 2). This is the one place our results diverge substantially rather than marginally. The source study reports peak $|\rho| < 0.19$ and peak MI < 0.04 nats across all six models and every layer, indistinguishable from its own random-direction baseline — its central claim is near-total dissociation between representation and verbalization. Our peak correlations are three to four times higher for both Qwen3-8B (0.639) and granite-4.2-30b (0.617), and visibly clear our own random-direction band. As reported in §4, we verified this is a shared-ground-truth confound rather than a bug: our direct-question probes are high-accuracy classifiers of ground truth (up to 100% for some Qwen3-8B probes), so correlating the internal probe score against the verbalized signal is close to mathematically the same computation as correlating it against ground truth directly, and coupling strength tracks probe accuracy almost linearly across our five probe phrasings.

The source study’s own robustness appendix (Appendix D) makes this mechanism visible from the other direction. When the authors replace their conservative 0–3 judge with a more permissive 0–100 judge, measured verbalization *rates* on post-trained Olmo3 checkpoints jump from 3–6% to 57–67% positive, purely from loosening the scoring rubric — yet peak coupling stays weak under *both* judges (Appendix G.3: peak $|\rho| \leq 0.17$, peak MI ≤ 0.15 nats even under the permissive judge). In other words, the source study already shows that verbalization-rate estimates are highly sensitive to how “verbalization” is operationalized, while coupling stays weak across that entire sensitivity range. Our result extends this same axis one step further: moving from an incidental free-form judge (their method, even at its most permissive) to a direct elicited yes/no question (our method) does not just shift the verbalization rate — it breaks the near-independence entirely, because a direct question answered accurately is, definitionally, close to the ground-truth label itself. Read together, the two studies suggest that “representation and verbalization are weakly coupled” is not a fixed fact about model internals; it is contingent on how permissively or incidentally verbalization is elicited and scored, with our result marking one endpoint of that spectrum and the source study’s 0–3 judge marking another.

Qwen3-8B’s steering profile, on the one directly overlapping model. Qwen3-8B is the only model tested in both studies, which makes this comparison unusually direct. The source study reports Qwen3-8B as its *cleanest* steering result of the four models tested in §4: a peak $\Delta J(\alpha, \ell) \approx 1.5$, concentrated in a narrow layer band, with negative α reliably increasing the judge score. Our own Qwen3-8B result is messier at the same nominal model: at moderate strength, *both* signs of α tended to increase mean `eval_signal` above baseline (e.g. layer 23: $\alpha \in \{-0.5, +0.5, +1.0\}$ all yield $\Delta = +0.5$), and full-strength eval-ward steering broke generation outright. We see two plausible, non-exclusive explanations, neither of which we can fully adjudicate at our sample size: (i) our binary rule-based outcome metric is coarser than the source study’s 0–3 judge and may be more sensitive to surface-level output degradation being misparsed as a signal shift rather than a genuine verbalization change; (ii) our 6-transcript steering subset is two to three orders of magnitude smaller than the source study’s 1000-prompt sweep, so a handful of outlier transcripts can dominate the reported mean. This divergence is disclosed rather than smoothed over precisely because it is the one place where a same-named model produces a qualitatively different steering profile across the two studies, and it is a natural target for a larger-scale follow-up.

A.5 Facets not attempted here

The source study’s fourth facet — evolution across the Olmo3 Base/SFT/DPO/Think training checkpoints — has no counterpart in this report at all, for a reason external to methodology:

none of our three Phase-2 models have publicly released intermediate training checkpoints, so the experiment is not constructible with the models available to us, independent of compute budget. We also did not attempt the source study’s judge-robustness sensitivity analysis (Appendix D: rescoreing every generation under a second, differently scaled judge) on our own probes, since our probes are already a rule-based direct-question design rather than an LLM-judge scoring free text, so the analogous robustness question — does the finding survive a different scoring rubric — does not apply in the same form; the closest analogue we do report is the Experiment 2 divergence discussion above, which addresses the same underlying concern (sensitivity of coupling estimates to how verbalization is measured) from the opposite direction.